

SUMMARY

Quantitative Researcher with modeling experience across vol forecasting, options pricing, crypto, and portfolio optimization

EDUCATION

Cornell University, College of Engineering, Ithaca, NY December 2025

Master of Engineering in Financial Engineering, GPA: 3.72

Relevant Coursework: ML, Optimization, Stochastic Calculus, Algorithmic Trading, Cryptocurrency, Derivatives

University of California Santa Barbara, Santa Barbara, CA June 2024

Bachelor of Science in Financial Mathematics & Statistics, GPA: 3.53

Relevant Coursework: Computer Science, Probability & Statistics, Stochastic Process, Data Science, Linear Algebra

TECHNICAL SKILLS

Software & Tools: Python (Pandas, NumPy, SciPy, Scikit-Learn, LightGBM, XGBoost), C++/C#, Java, Julia, R, SQL, MATLAB, Git, PyTorch/TensorFlow, React, AWS, CVXPY, Shell-Scripting

PROFESSIONAL EXPERIENCE

Research Analyst, *Indxx Inc.*, Miami, FL January 2026 - Present

- Designed and productionalized an intraday volatility forecasting & index-calculation pipeline for a live vol-controlled index, including data ingestion, model serving, automated signal generation, and a React dashboard for rapid prototyping
- Outperformed industry-standard HAR volatility model by 9.6-36.9% (out-of-sample MSE) via ML-based forecasting, validated across 10+ years of backtesting and five U.S. equity benchmark indices
- Developed and validated two European options pricing models (Black-76+SVI, Heston) with leakage-free same-day K-fold CV; cut pricing MAE 71%, implied-vol RMSE 73% vs two flat-vol baselines (Black-76, Black-Scholes-Merton)

Alpha Researcher, *Trexquant Investment LP*, New York, NY August - December 2025

- Engineered 200+ predictive features from 400 opaque alpha signals via IC-based screening, LightGBM importance, GA-driven interaction mining, and cross-sectional transformations; controlled multicollinearity via correlation-based pruning
- Combined alpha, financial, and NLP-derived news-embedding features into LightGBM model achieving 0.041 OOS IC; ranked 1st among competing teams with private leaderboard score of 0.072 and won \$5000 bonus

Quantitative Analyst Intern, *Shepherd Ventures*, San Diego, CA (Remote) April - December 2025

- Constructed regime-adjusted S&P 500 portfolio using K-Means, beta normalization, Holt estimator, and Mean-Variance Optimization; achieved 86% downside protection vs. SPY across 12 historical crisis windows
- Enhanced recovery strategies post-market downturns by developing normalized recovery metrics and implementing A/B testing to support forecasting, financial analysis, strategic planning, and business management in portfolio risk management

Quantitative Research Assistant, *Cornell University*, New York, NY February - December 2025

- Conducted regression modeling and statistical analysis to 6,069 stocks and 12 sector ETFs across 240 months, identifying a significant positive correlation between market cap and stock-ETF clustering patterns
- Authored an academic paper translating quantitative findings on industry drift and clustering into actionable strategies

Asset Management Analyst Intern, *Nomura Orient International Securities Co., Ltd.*, Shanghai July - September 2023

- Improved CAPM and DCF prediction accuracy by 20% via systematic data preprocessing and automated outlier detection
- Delivered a comprehensive report with actionable insights on Microsoft's AI Copilot and Kingsoft's WPS AI

KEY PROJECTS

Limit Order Book (LOB) Execution Strategy Project, *Cornell University*, Ithaca, NY March - May 2025

- Beat TWAP by up to 80% via adaptive execution strategy using OFI, MACD, and bid-ask spread signals from real-time LOB data, validated through statistical robustness testing

Regime-Conditional Portfolio Optimization Project, *Cornell University*, Ithaca, NY January - May 2025

- Achieved 0.91 Sharpe ratio in multi-asset ETF portfolio via regime classification, rolling estimators, and convex optimization with dynamic risk calibration, automating model recalibration and allocation parameter selection